LED Control System

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B.E. CSE (Cyber Security), 3rd Year
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Introduction

- This project aims to build an efficient LED control system using an Arduino microcontroller. It helps in understanding basic circuit building and embedded programming.

Objective

- To develop a simple, low-cost system that allows LEDs to be controlled using code and demonstrates basic logic flow in embedded systems.

Components Used

- - Arduino Uno
- - LEDs (Red, Green, Blue)
- - Resistors (220 ohms)
- - Jumper Wires
- - Breadboard
- - USB Cable for Power

Working Principle

- The Arduino sends output signals to the LEDs through digital pins. Each LED lights up one after another with a time delay, creating a sequence pattern.

Circuit Description

- Three LEDs are connected to digital pins 9, 10, and 11 respectively. Each pin is programmed to turn the LEDs on and off in a cycle with a 1-second delay.

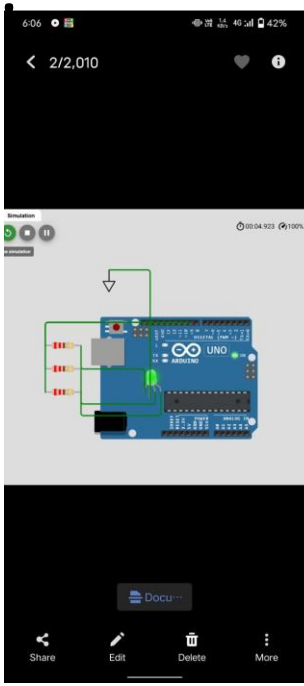
Microcontroller Used

- We use the Arduino Uno microcontroller, which is an open-source electronics platform based on easy-to-use hardware and software. It is ideal for prototyping.

Flow Chart

- Start -> Initialize pins -> Turn ON Red -> Delay -> Turn OFF Red, Turn ON Green -> Delay -> Turn OFF Green, Turn ON Blue -> Delay -> Repeat

Block Diagram

- Refer to the block diagram where the Arduino


cted to 3 LEDs with appropriate
. It shows the flow from code to
e.

Program Code

- See the embedded code. It cycles through the

sing `digitalWrite()` and

in the Arduino IDE.

Implementation

- 1. Connect the LEDs to the Arduino board.
- 2. Open Arduino IDE and upload the program.
- 3. Power the board and observe the LED sequence.
- 4. Debug if needed.

Advantages

- - Low Power Consumption
- - Easy to Program and Use
- - Inexpensive Components
- - Scalable for Larger Projects

Applications

- - Smart Home Lighting
- - Indicator Systems
- - School and College Projects
- - Industrial Light Controls

Challenges Faced

- Initially faced wiring issues and incorrect pin mappings. Solved by reviewing pin definitions and ensuring correct connections.

Future Scope

- Can be expanded with IoT integration using Wi-Fi modules like ESP8266, or controlled using mobile apps and voice assistants like Alexa.

Conclusion

- This project enhanced our understanding of embedded systems. We thank our mentor Mr. Santosh and JJ College for the guidance and support.